

Autonomous Mars Rover: a propositional logic agent

BCA301-5 Artificial Intelligence · AI Express Hackathon · **Group 6** · Track 2: Autonomous Mars Rover (Unit 3, Propositional Logic Agent), with A* replanning over the proved-safe squares
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Repository: <https://github.com/supprith/mars-rover-logic-agent>

1. PEAS framework

Performance	+1000 for docking at the lander with the sample, -1000 for losing the rover, -1 per action, -10 for firing the containment charge. Secondary measures logged every run: path cost in moves, A* nodes expanded and generated, number of replans, clauses held, resolution steps, reasoning time.
Environment	6x6 survey grid of safe terrain, unknown hazards and one radiation zone. Each square outside the lander and its neighbours holds a hazard with probability 0.16, plus one radiation zone and one sample cache. Generation retries until the sample is reachable without crossing a hazard. Partially observable (readings are local only), deterministic, sequential, static, discrete, single agent.
Actuators	Drive to an orthogonally adjacent square (north, south, east, west), Collect the sample, Neutralise the radiation zone along a bearing with the single containment charge, Dock at the lander.
Sensors	Hazard warning (an unknown hazard is adjacent), radiation alert (the radiation zone is adjacent), sample beacon (the cache is on this square), neutralise confirmation, bump (drove into the edge). Nothing else is visible: the rover never sees a square it has not parked on.

2. Core algorithmic formulation

State space	Rover state is (position, knowledge base, has_sample, has_charge, source_active). The planner searches a smaller graph: vertices are the squares currently proved safe or already driven on, minus squares proved to hold a hazard; edges join orthogonal neighbours. That vertex set changes after every reading, which is what forces the replan.
Initial state	Rover at the lander (0,0) facing east, charge unused, no sample. KB holds only Visited(0,0) plus the adjacency facts for the grid. Nothing about the hazards, the radiation zone or the sample cache is given.
Goal test	Mission: Dock executed at (0,0) while has_sample. Search: n = target square, where the target is the cheapest safe square not yet surveyed, or (0,0) once the sample is aboard.
Path cost	$g(n)$ = number of moves from the current square, one unit per edge, so $g(n)$ is the length of the path in the search graph.
Heuristic	$h(n) = x_n - x_{goal} + y_n - y_{goal} $, $f(n) = g(n) + h(n)$. Admissible: every move changes Manhattan distance by exactly 1, so no path can be shorter than $h(n)$. Consistent: for adjacent n, n' , $h(n) \leq c(n, n') + h(n') = 1 + h(n')$, so A* with a closed set returns an optimal path and never reopens a node.
First-order rules	$Visited(c) \Rightarrow NoHazard(c)$ · $Visited(c) \Rightarrow NoRadiation(c)$ $Visited(c) \text{ and } NoWarning(c) \text{ and } Adjacent(c, n) \Rightarrow NoHazard(n)$ $Visited(c) \text{ and } NoAlert(c) \text{ and } Adjacent(c, n) \Rightarrow NoRadiation(n)$ $NoHazard(c) \text{ and } NoRadiation(c) \Rightarrow Safe(c)$ · $RadiationSealed \text{ and } Cell(c) \Rightarrow NoRadiation(c)$ Applied by forward chaining to a fixpoint, with unification against the ground adjacency facts.
Propositional axioms	$W_{x,y} \Leftrightarrow \text{OR over } n \text{ in } Adj(x,y) \text{ of } H_n$ and $A_{x,y} \Leftrightarrow \text{OR over } n \text{ in } Adj(x,y) \text{ of } Z_n$, where W is the hazard warning, A the radiation alert, H an unknown hazard and Z the radiation zone. Added in CNF the first time a square is seen, together with the unit clauses not H_c , not Z_c for every square driven on and the observed W_c or not W_c .
Entailment	KB entails a query iff KB together with the negated query is unsatisfiable. Tested by resolution refutation with a set-of-support strategy seeded on the negated query, a relevance filter that drops clauses whose symbols lie more than two moves from the query square, a maximum clause length of 8, and a 2500 step budget. Exhausting the budget returns "not proved", so the agent can only become more cautious, never less.
Risk when no proof exists	$P(H_c \mid \text{evidence}) = \frac{[\text{sum of } w(m) \text{ over the models that satisfy the evidence and set } H_c \text{ true}]}{[\text{sum of } w(m) \text{ over every model that satisfies the evidence}]}$, where $w(m)$ is the product of p over the true hazard variables and $(1-p)$ over the false ones, $p = 0.16$. Enumerated over the unproved frontier variables only. The rover drives onto the minimum if it is at or below 0.25, otherwise it returns to the lander and docks.

3. Complexity analysis, theory against measurement

Component	Theoretical	Observed over 400 maps
A* per search	$N = n^2 = 36$ squares, branching 4. Time $O(N \log N)$ with a closed set and a consistent heuristic, worst case $O(b^d) = 4^{10}$ without one. Space $O(N) = 36$.	3.5 nodes expanded per search, 67.8 per map over 19.3 replans; 117.4 generated
Forward chaining	Adjacency is bounded at 4, so the rules have $O(n^2)$ ground instances. $O(n^2)$ work per pass, $O(n^4)$ to reach the fixpoint.	64 facts derived per map, 259 ground facts held at the end
Resolution	Propositional resolution is $O(2^m)$ in the worst case for m symbols. Here $m = 4n^2 = 144$, cut to roughly 52 by the relevance window and then capped by the 2500 step budget, so each query is bounded.	106.1 queries per map, 1298 steps per query against the 2500 cap
Model counting	$O(2^k)$ over k unproved frontier variables, skipped entirely when $2^k > 65536$.	k stays in the range 2 to 9 in practice, so under 512 assignments per call
Whole episode	Turns bounded by the 250 step cap; memory $O(n^2)$ facts plus $O(n^2)$ clauses	21.1 turns, 19.3 moves, 190.4 clauses, 371.4 ms of reasoning per map; 94.2% of rovers survived, 64.5% recovered the sample, mean score 564.2

Measurements reproduce with `python run.py --benchmark 400 --seed 0` on Python 3.11. The recorded demo run is `python run.py --seed 114 --delay 2400`: resolution proves a hazard at (2,0), model checking pins the radiation zone at (1,1), 19 turns, 16 moves, 47 nodes expanded, 34 resolution queries with 2 proofs, final score 971.